

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1504

COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Manipulation (Level 2)

Question Count: 5

Total Marks: 25

Code of Conduct

I **Edrin J S** (Name / Reg No) **192524041** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *Edrin J S*

Date: 29 July 2026 **Batch:** 2025

Criteria	Understanding of Scenario (1)	Identification of Risks / Issues (1)	Application of Concepts (1.25)	Decision Justification (1)	Clarity (0.75)	TOTAL (5)
Weightage	20%	20%	25%	20%	15%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Signature of the Faculty

Total Marks Awarded:

SCENARIO-BASED QUESTIONS — ANSWERS

1. A financial institution plans to deploy a private cloud with strict compliance requirements. Recommend an appropriate cloud datacenter architecture and explain how virtualization and isolation mechanisms can ensure security and regulatory compliance.

Recommended architecture: A dedicated private cloud, either on-premises or hosted on a single-tenant infrastructure, is most appropriate given the strict compliance requirements. It should use a multi-tier datacenter design with clearly segregated compute, storage, and network layers, dedicated hardware pools for the most sensitive workloads, and a management layer providing centralized policy enforcement, logging, and monitoring. How virtualization and isolation ensure security and compliance:

- The hypervisor enforces strict logical isolation between VMs, ensuring departments or workloads cannot access each other's data even while sharing hardware.
- Network micro-segmentation and VLANs separate traffic for regulated workloads (e.g., transaction processing) from general-purpose systems.
- Role-based access control (RBAC) and multi-factor authentication restrict administrative access to virtualization management consoles.
- Encryption of data at rest and in transit, combined with hardware security modules (HSMs) for key management, protects sensitive financial data.
- Comprehensive audit logging and immutable log storage support regulatory audits (e.g., PCI-DSS, SOX) by providing traceable records of every access and change.

2. A SaaS provider experiences inconsistent application performance due to noisy neighbor effects in a virtualized environment. Analyze the root cause and propose resource allocation and hypervisor-level solutions.

Root cause: The noisy neighbor effect occurs because multiple VMs share the same underlying physical CPU, memory, storage I/O, and network bandwidth. When one VM consumes a disproportionate share of these resources (e.g., a batch job spiking CPU usage), co-located VMs experience contention and degraded, inconsistent performance — this is worsened when the hypervisor lacks strict resource limits or fair scheduling.

Proposed solutions:

- Resource allocation: Set CPU and memory reservations, limits, and shares per VM so no single tenant can exceed its fair allocation; apply storage IOPS throttling and network bandwidth shaping (QoS policies) to cap heavy consumers.
- Hypervisor-level solutions: Use fair-share or proportional scheduling algorithms, enable CPU pinning and NUMA-aware VM placement to reduce cross-node contention, and use dynamic resource schedulers to automatically migrate VMs away from overloaded hosts.
- Isolate consistently heavy tenants onto dedicated hosts or resource pools, and continuously monitor utilization to proactively rebalance workloads before performance is affected.

3. An organization wants to migrate legacy monolithic applications into a Service-Oriented Architecture (SOA) in the cloud. Explain the migration strategy and how SOA improves scalability and maintainability.

Migration strategy:

- Assess the monolith and identify logical functional boundaries (e.g., billing, authentication, reporting) that can be extracted as independent services.
- Apply the strangler fig approach: incrementally extract and replace individual modules with standalone services while the monolith continues running, rather than a risky one-time rewrite.
- Define clear service contracts and interfaces (REST/SOAP APIs with WSDL or OpenAPI specifications) so each service can be developed and deployed independently.
- Introduce an API gateway and service registry to route requests and manage service discovery, and use an integration layer (e.g., an enterprise service bus or message broker) to bridge legacy and new components during transition.
- Decouple data by giving each service its own data store where feasible, and adopt CI/CD pipelines to test and deploy each migrated service independently.

How SOA improves scalability and maintainability: Because services are loosely coupled and independently deployable, each one can be scaled horizontally based on its own demand rather than scaling the entire application. Teams can update, test, and deploy individual services without affecting the rest of the system, faults are isolated to

individual services rather than bringing down the whole application, and common functionality can be reused across multiple services — all of which reduce long-term maintenance effort compared to a tightly coupled monolith.

4. A government agency is implementing identity federation across multiple departments using cloud services. Explain how identity management, authentication, and privacy policies should be designed to ensure secure access.

Identity management: A centralized Identity Provider (IdP) should be established using federation standards such as SAML, OAuth 2.0, or OpenID Connect, allowing each department to retain its own user directory while trusting a shared federation layer for single sign-on (SSO). This avoids duplicating identities across systems and centralizes policy enforcement.

Authentication: Access should require multi-factor authentication (MFA), especially for sensitive systems, along with strong password and credential policies. Secure token-based authentication (e.g., SAML assertions or OAuth access tokens) should be used for cross-department access instead of sharing raw credentials, and tokens should have limited lifetimes with proper session management.

Privacy policies: Access should follow the principle of least privilege, using attribute-based or role-based access control (ABAC/RBAC) so each department only receives the identity attributes it needs (data minimization). All identity data should be encrypted at rest and in transit, access should be logged and regularly audited, and data-sharing agreements between departments should comply with applicable privacy regulations, with periodic access reviews to revoke unnecessary permissions.

5. A cloud provider wants to implement real-time user presence detection for collaborative applications. Discuss how presence services can be integrated into cloud architecture while maintaining user privacy and system scalability.

Architectural integration: Presence updates should be handled through an event-driven, publish-subscribe model using persistent connections (WebSockets) or lightweight protocols (MQTT), so status changes are pushed to subscribers in real time instead of relying on inefficient polling. Presence state should be stored in a fast in-memory data store (e.g., Redis) rather than a traditional database, and presence servers should be horizontally scalable behind a load balancer, with users sharded across nodes and a distributed message broker (e.g., Kafka) used to buffer and distribute updates during peak load.

Maintaining scalability: Geographically distributed edge servers can reduce latency for global users, caching frequently accessed presence data reduces backend load, and asynchronous processing prevents presence updates from blocking core application functionality.

Maintaining privacy: Users should be able to control their visibility (e.g., online/away/invisible modes) and only share detailed presence information with authorized contacts. Presence data should be encrypted in transit, retained only as long as necessary, access should be rate-limited to prevent scraping, and audit logging should track who accessed presence information to ensure accountability.